

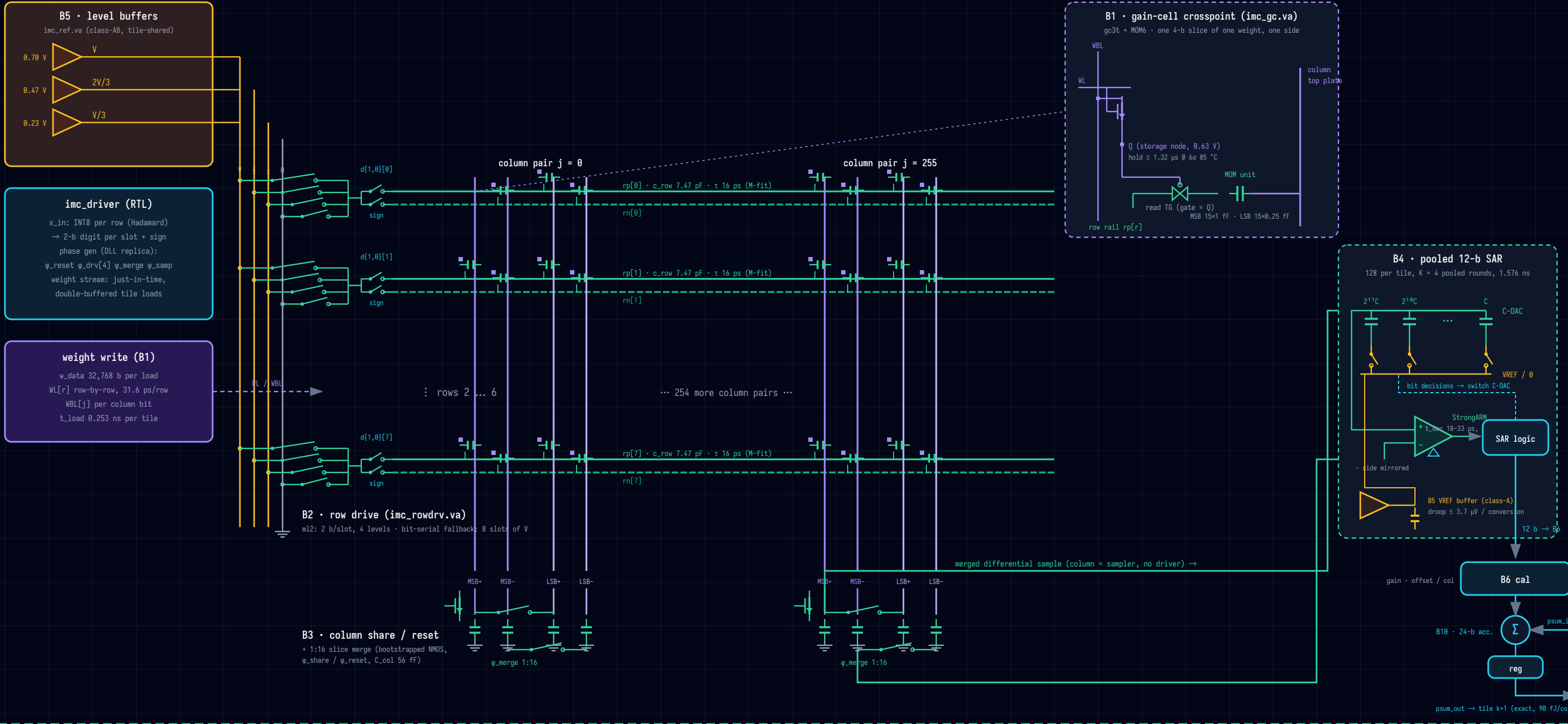
OVERVIEW · one tile pass of a GEMM / GEMV

All blocks of tile k. The next six pages walk one pass of the operation phase by phase; active circuitry stays bright, the rest is dimmed.
GEMM (prefill): one weight load serves all 512 prompt tokens, one pass per token. GEMV (decode): one weight load serves every concurrent stream (B = 587 at the ARCH point), one pass per stream.

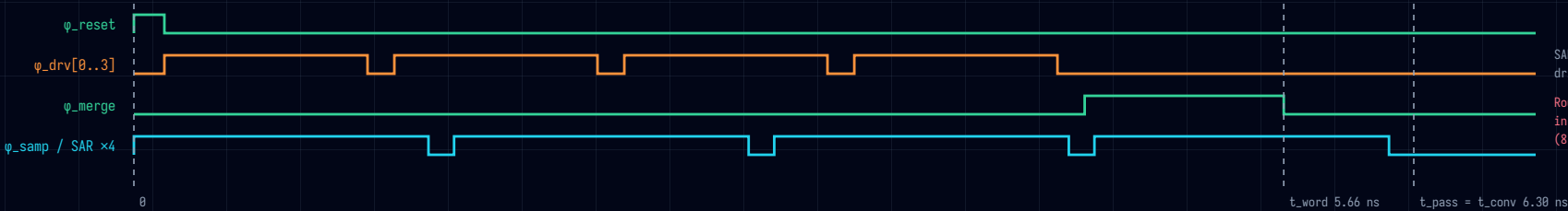
SYSTEM · one die (ARCH: 100 mm², 3,181 tiles)



TILE k · 18,962 μm² · one pass = 2,048 MACs in t_{pass} = 6.30 ns (conversion-bound) · VDD_A 0.7 V, VDD_L 0.5 V



ONE PASS · phase timing (imc_driver)



SAR rounds of pass n overlap the drive slots of pass n+1 (ping-pong)
Round 2: the 4-level drive cannot settle in a 1.132 ns slot → bit-serial fallback (8 slots of V, constant-charge coding)

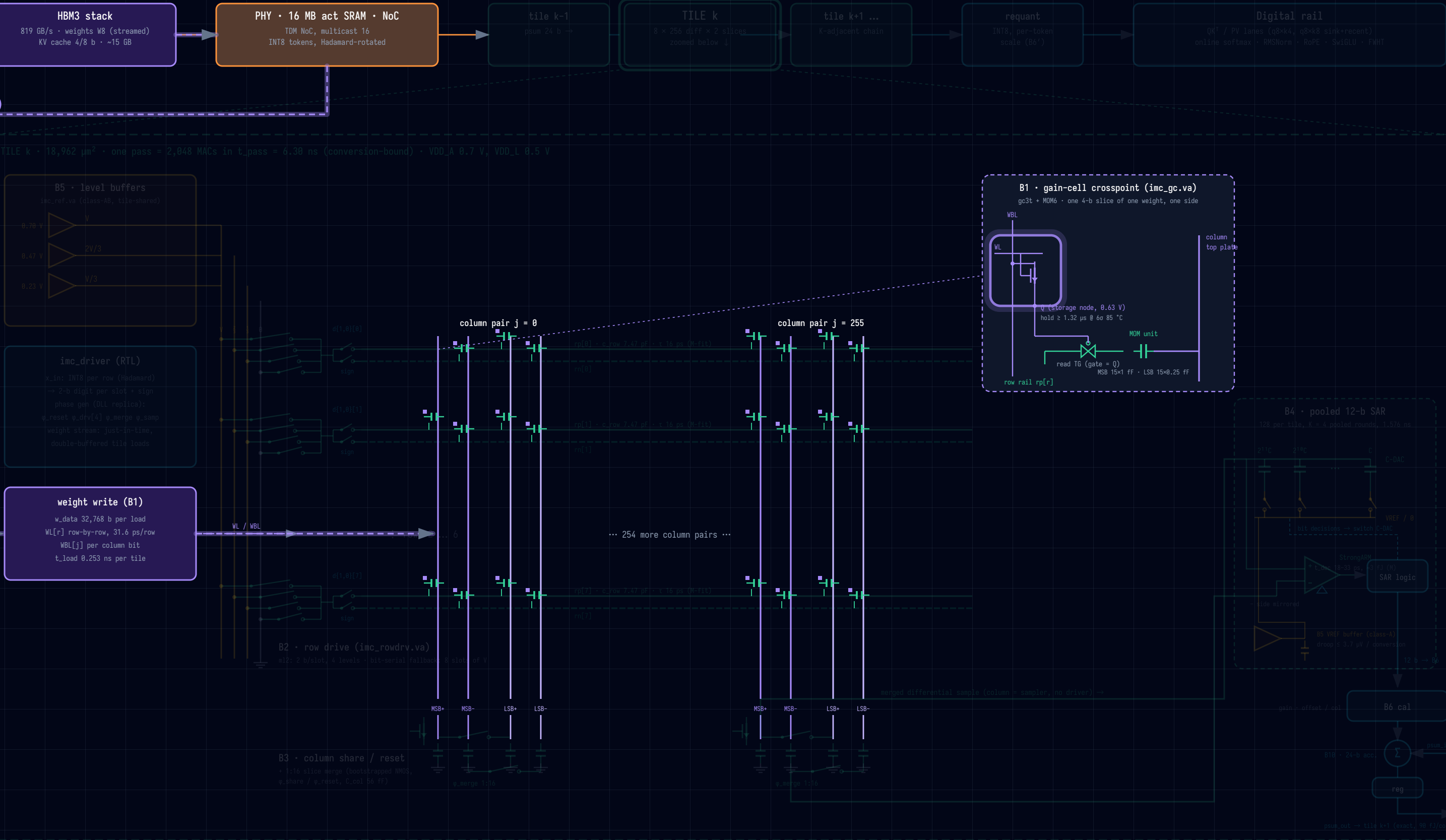
LEGEND

- analog signal path
- weights / write path
- references & level nets
- digital & clocks
- streams from HBM / NoC
- open risk (round 2)
- MOM / C-DAC capacitor
- switch
- transmission gate
- buffer / level driver
- StrongARM comparator
- accumulator

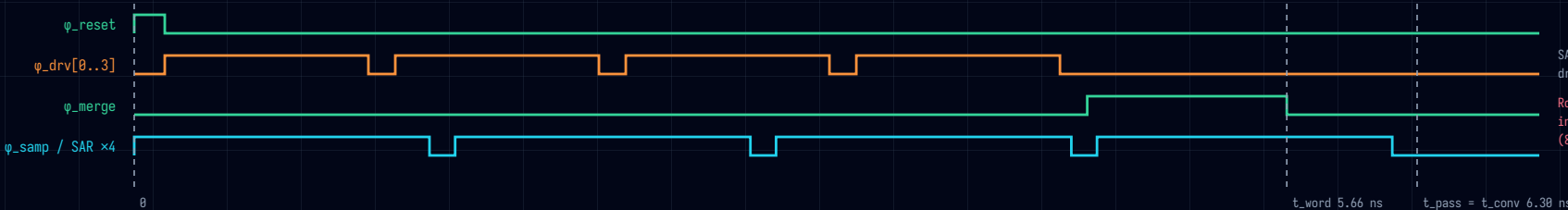
PHASE 1 • weight load (streamed from HBM, ahead of the activations)

Weights stream HBM → PHY/NoC → the tile's write port and are written row by row into the gain cells: WL[r] high puts the bit on WBL[j] onto the storage node Q (31.6 ps per row).
One tile load = 32,768 bits in 0.253 ns of array time, double-buffered so it hides under the previous pass. Write energy ≤ 0.7 fJ/bit (0.52 fJ per '1' measured).
GEMM: written once, reused by 512 token vectors. GEMV: written once, reused by every concurrent stream; HBM bandwidth (weights + KV) bounds how often loads can happen.

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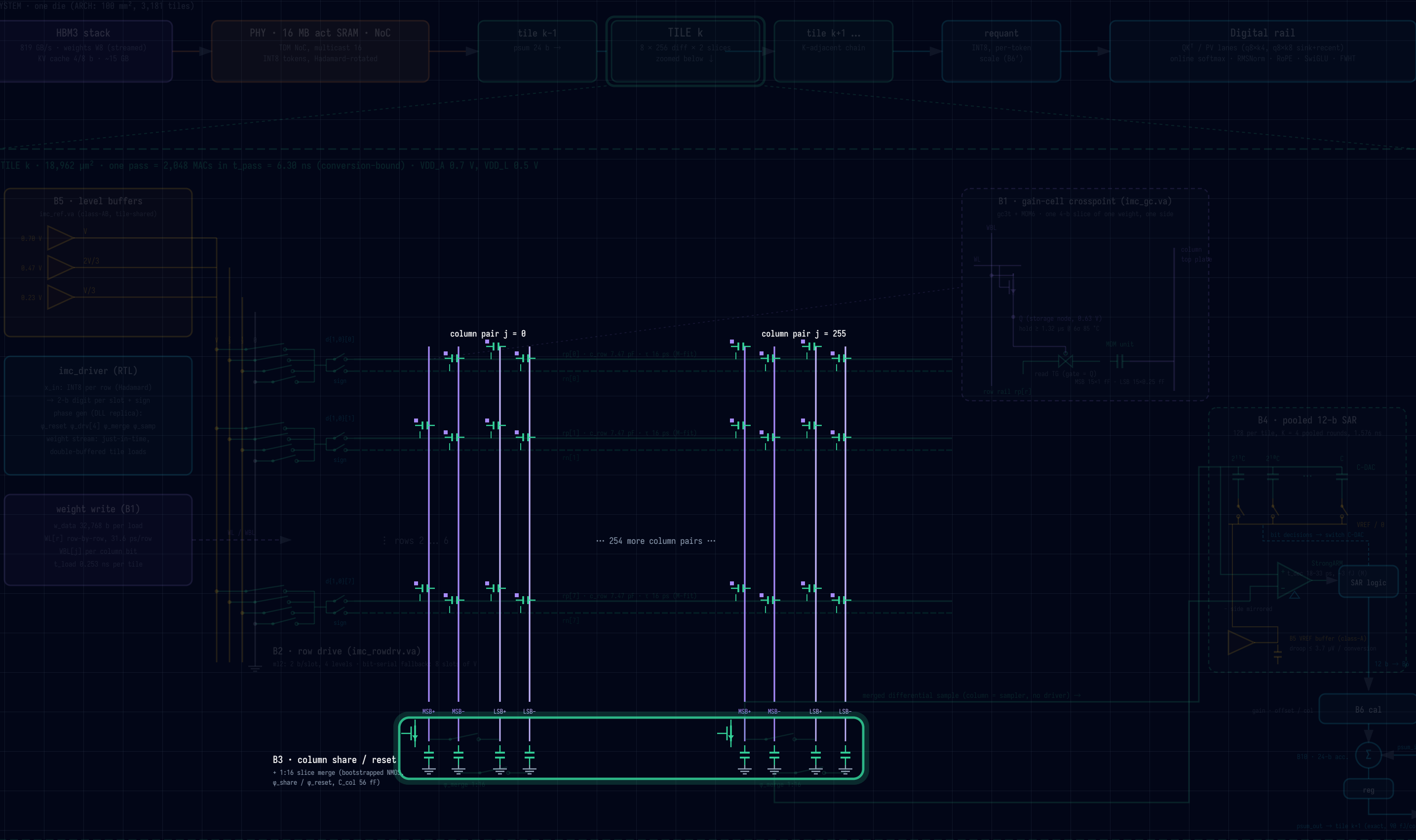


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PHASE 2 • reset $\varphi_{\text{reset}} 0 \rightarrow 0.15 \text{ ns}$

φ_{reset} : the bootstrapped NMOS switches clear every column top plate and C_col (56 fF) to the common level; the row rails sit at 0.
This removes the previous pass's charge so the column starts empty. Switch $\tau \leq 64.5 \text{ ps}$; the gate ($\sim 1.3 \text{ V}$ bootstrapped) must stay inside the ASAP7 V_{GS} limit.



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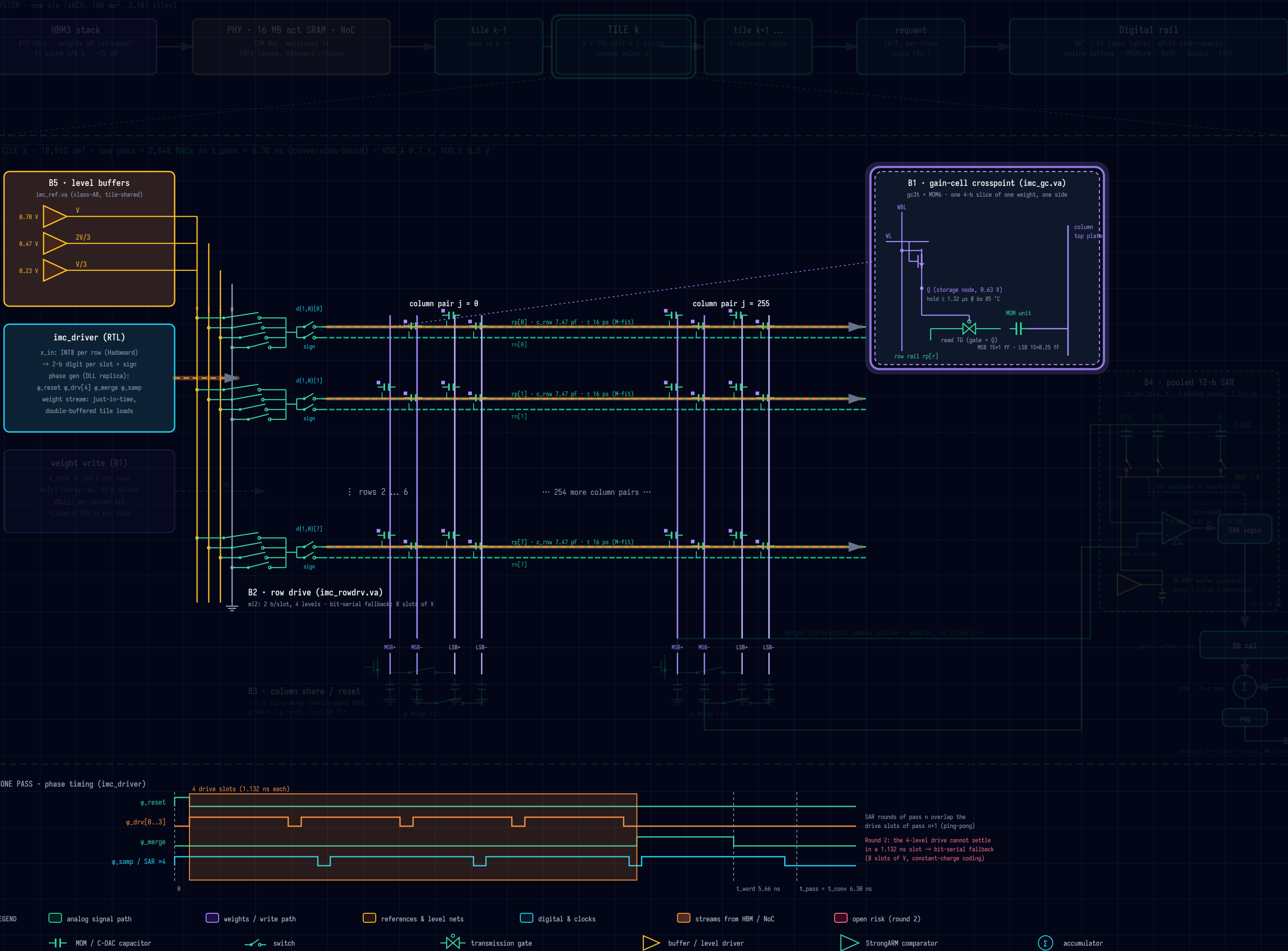
buffer / level driver

StrongARM comparator

accumulator

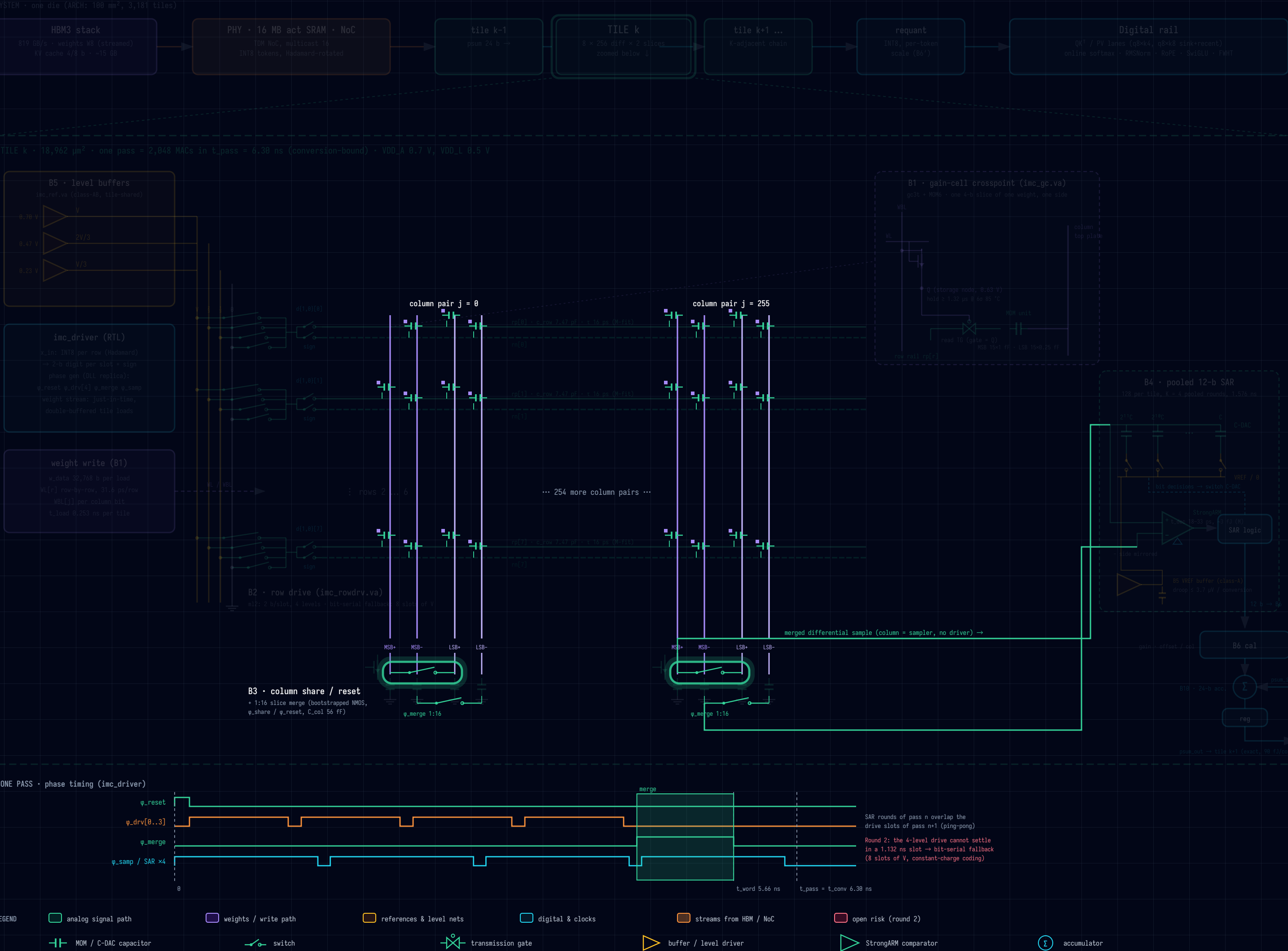
PHASE 3 • drive slots: multiply and accumulate in charge $\varphi_{\text{drv}}[0..3]$ 0.15 → 4.68 ns

Each slot s (LSB first) turns 2 activation bits per row into one of 4 rail levels (0, $V/3$, $2V/3$, V) on the row's + or - bottom-plate rail (the sign picks the rail). Every MOM unit whose gain cell holds a '1' couples that rail step into its column: multiply = charge $Q = C_{\text{unit}} \Delta V$; accumulate = all 8 rows' charge summing on the column (KCL). Each new slot shares with the charge already on the column, so it ends at $v = 2^{-8} \cdot \sum 2^j \cdot v_j$ (charge radix): one conversion per column for the whole 8-bit input. Round 2: the 4-level rails cannot settle in a 1.132 ns slot → bit-serial fallback (8 slots of 0/V with constant-charge coding).



PHASE 4 • slice merge φ_{merge} 4.68 $\rightarrow$ 5.66 ns

φ_{merge} : the LSB-slice column shares charge with the MSB-slice column at a 1:16 ratio, so the merged sample is $y = 16 \cdot y_{\text{MSB}} + y_{\text{LSB}}$.
The 8-bit weight is rebuilt in charge, not in digital: no slice recombination logic. Target ratio error $\leq 0.1\%$ (unverified: verification V2).



PHASE 5 • conversion φ_{samp} + 4 pooled SAR rounds 6.30 ns (overlaps the next pass's drive)

The column itself is the sampler (no driver). 128 SARs per tile run 4 pooled rounds ($K = 4$) of 1.576 ns: the C-DAC bits switch between VREF and 0, the StrongARM decides each bit (18–33 ps, ~3 fJ, measured), and the SAR logic builds a 12-b code. The reference droops $\leq 3.7 \mu\text{V}$ per conversion. This phase sets $t_{\text{pass}} = 6.30 \text{ ns}$ and is the top risk: 12 b at $\geq 635 \text{ MS/s}$ and $\leq 254 \text{ fJ}$ is 5–8 dB past the published envelope (round 2 offers a residue-amp SAR).



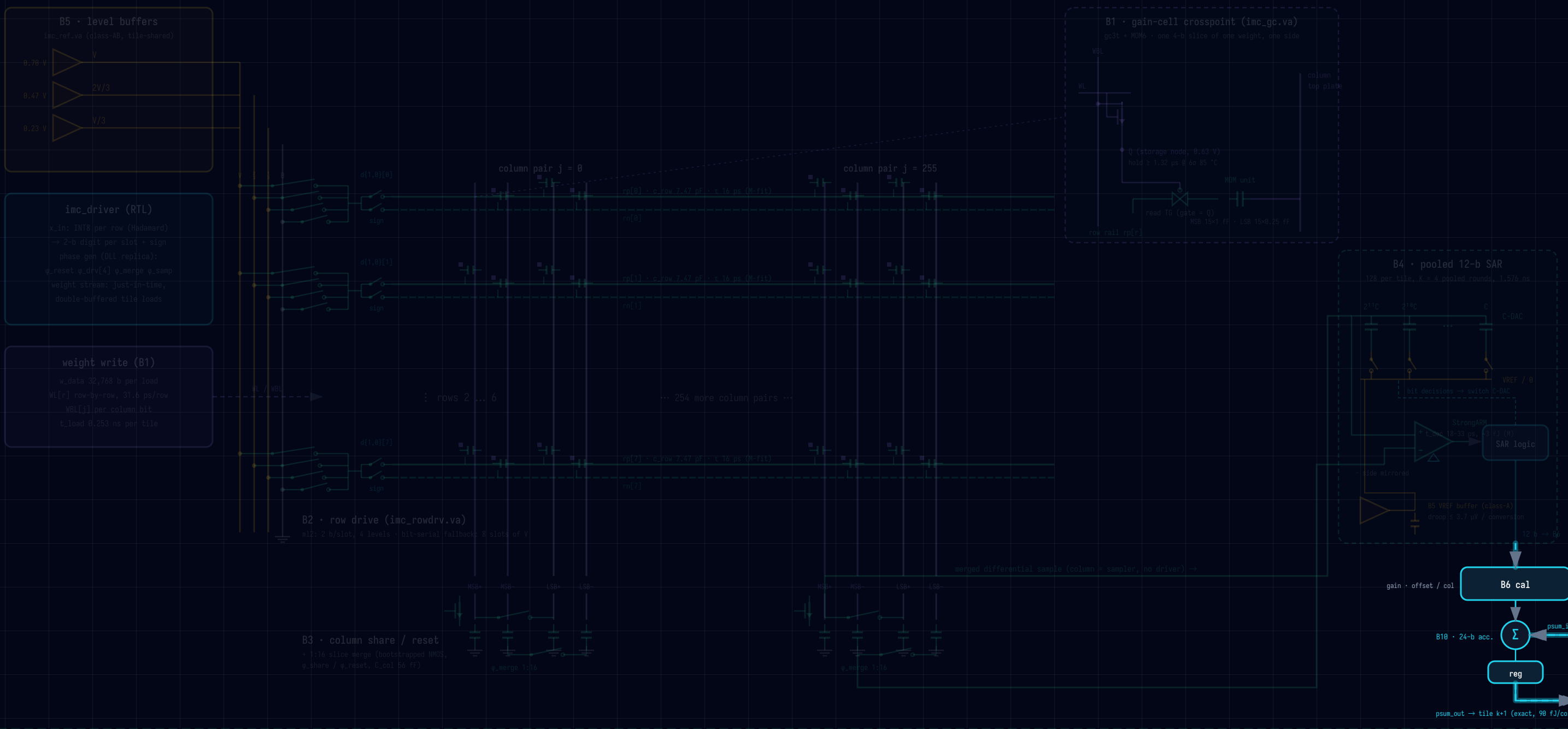
PHASE 6 • calibrate, accumulate, hand off one hop per t_pass (pipelined)

B6 applies each column's gain and offset; B10 adds the code to the 24-b partial sum arriving from tile k-1 and passes it on to tile k+1 (exact, ~90 fJ per column).
The K-chunks of one output run down K-adjacent tiles; only the completed sum leaves the chain, is requantized to INT8 with the per-token scale, and goes to the digital rail.
GEMM: the chain delivers one output vector per token per pass. GEMV: one per concurrent stream; the digital rail then runs attention against the 4/8-b KV cache.

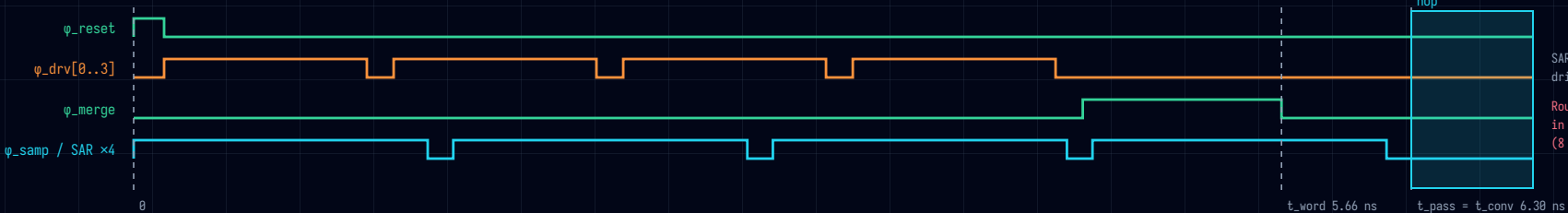
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